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INFO I-101

Paul

Brief Biography

Dr. Shirley Jackson, was born on August 5 th , 1946. During her childhood, she took a particular interest in science and would often do science experiments. She would go on to be the first black woman to earn a PhD in physics from MIT and would follow that up by becoming a leading expert in theoretical physics. She would go onto work at Bell Laboratories, "her research there contributed to the inventions of the touch-tone telephone, fiber-optic cables, caller ID, and call waiting." (Ramanathan, 2023). Following this, in 1995, she was appointed the head of the Nuclear Regulatory Commission, and then, finally, "she became the eighteenth president of Rensselaer Polytechnic Institute." (The HistoryMakers, 2006).

Underrepresentation

Dr. Shirley Jackson is one of strongest examples of underrepresentation, in MIT she was, "among the first African American students to attend MIT, and in her undergraduate class she was one of only two women." (The HistoryMakers, 2006).To follow that up she would go on to be the first woman to earn a PhD in physics from MIT. Beyond that, she excelled in an environment that was institutionally hostile against her, as she was forced to attend segregated schools growing up and was told by her school advisor that "colored girls should learn a trade." (Success Academy 2019). She overcame these structural hurdles by believing in herself and pushing towards her goals no matter what. By now it is pretty safe to say that Dr. Shirley Jackson, who was the head of The U.S. Nuclear Regulatory Commission, Principal of the Rensselaer Polytechnic Institute, the co-chair of the President's Intelligence Advisory Board, and recipient of the National Medal of Science, has overcome the structural prejudice against her and excelled despite her circumstances.

Contribution

While working at Bell Laboratories, “her research there contributed to the inventions of the touch-tone telephone, fiber-optic cables, caller ID, and call waiting.” (Ramanathan, 2023).

Future Promise

Early research by Dr. Shirley Ann Jackson in telecommunication enabled modern forms of communications like fiber optic and caller ID, which also help define future technologies today. Her research laid the groundwork for future advances in 6G networks, which will enable faster global communication, virtual reality, and smart city networks. Her physics training also ties into the growing field of quantum communication, which provides ultra-secure transmission of information. As technology progresses, the legacy of Jackson will keep setting the pace for a faster, smarter, and more accessible world to connect worldwide.

Selection

Our decision was between Hedy Lamar and Dr. Shirley Jackson, both were innovators who experienced underrepresentation and had to overcome the odds to have been incredibly influential on technology. For context Hedy Lamar during World War 2, used her passion for science to help her invent a frequency-hopping signal that prevents torpedoes from being jammed or tracked. The tech she invented is still used in Bluetooth and Wi-Fi, being an important component to it. However we decided against her being our innovator for two main reasons, one we felt that she would be a common pick. Secondly, when we looked up Dr. Shirley Jackson, a different person came up, this solidified in our minds that Dr. Shirley Jackson was truly underrepresented.

Media Assets

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